



Parametric vs. Non-Parametric Transcript

Hi everyone, let's get into details of parametric and non-parametric methods. We actually call them as parametric and non-parametric distributions and parametric and non-parametric statistical methods as well. There are various different ways in which we can define this, but we'll have to first understand what exactly is the meaning of parametric, right, or non-parametric.

When I say parametric, it involves the term parameter in it and just before this video, you learned about a normal distribution which had two parameters, mean and standard deviation. So a parametric distribution is about learning about the parameters of population. Let's say, I say the population of India, the height of the entire population of India can be measured with normal distribution.

Now when I say normal distribution, we can say that the average height of people living in India is 5.5 with a standard deviation of let's say 3, right, so majority of the people are in between let's say 2 to 8 feet, that's what it means. Now these two parameters define the entire shape of the distribution, okay. So parametric methods are all those methods which do a lot of assumption about our parent or population distribution, okay.

When I say population distribution, you also understand that we have a population, entire statistics revolves around taking a sample from a population and then inferring about the population using that specific sample. So let's say in India you have billions of people. Now if you have to make and study about different types of people living in India, you cannot go ahead and ask each and every individual who is living in India, it will take a lot of time plus also a lot of resources, right.

So what you do, you take a sample and then you study that sample, infer the results out of that sample and then portray it onto the entire population and that's what we use statistics for, right. Now in that entire process, we also assume a lot of things. We assume that the entire population follows a specific type of distribution and that's when we start talking about parametric and non-parametric.

When we make that assumption, that assumption or the distribution or the method, those types of things are called as parametric. So assumptions, whenever we involve assumptions, whenever we assume about anything about the population, those methods, those techniques are called as parametric. While when we do not assume about anything about the population, then those methods are called as non-parametric.

Think of this as a very simple analogy. Let's say you have a tailored suit, right. Now that tailored suit can be specifically fitted on a specific type of individuals, right.



It is not there for every individual. So parametric distribution is like a specific type of suit is made for a specific type of people, right. It cannot fit and it cannot go on and fit the entire range of people that we have.

So those are non-parametric. Now there are a lot of good and bad about parametric and non-parametric test which will be going into details in the list, last sections of the videos and all of those things. But right now, our idea is to get an initial understanding of what these things are, right.

So parametric distribution, they are really the specialist, especially crafted suit if you could relate to the analogy that I drove, right. So in that case, you have a lot of statistical powers. More likely you detect a true effect with one exist and let's say the cons are, it is very sensitive.

Now the other thing which is non-parametric, they are distribution free. They do not take any assumption about the distribution, but they are not very reliable. When I say reliable as in, they take less statistical power, which is one of the cons that they have.

But the pro side is they are very robust. Let's say you can take any data and then perform any statistical test which is non-parametric and get the results out of it. One very key distinguishing factor which we have between parametric and non-parametric is when you have ordinal data, let's say there is a competition wherein someone has to win first, second and third prize.

Now if we have different types of digits judging the first, second and third prize, in that case you can take a statistical test which is non-parametric because it works perfectly fine when you have ordinal data. So how do you decide which type of test could be used at what scenario? The very first thing is start with your data. If your data is ranked or ordinal, you have ranks in your data like first, second, third, high, medium, low and so on, you can very well use non-parametric.

But if you do not have that, you will have to use parametric. Within that, if your data is normally distributed, the type of distribution that we studied before this, if your data is normally distributed, you use parametric and then if your data is normally distributed, then use non-parametric. That's the entire assumptions we take and this is the entire flow that we have.

Now all the tests that we will be studying in this entire course like t-test, f-test and later on we will also study about ANOVA, they are all parametric methods because they take very hard assumption of the entire distribution following or the population distribution following normal distribution. While there are a lot of non-parametric tests which are there which we will be studying in the later section of the class, but the idea is to give you an introduction to all of these parametric tests which make an assumption



about the distribution or the assumption about what type of relationships are there and so on. These are called as parametric because there is an assumption involved.

While the other type of test wherein no assumption is involved, nothing at all. In those cases, we call these tests as non-parametric. As we were going through parametric and non-parametric methods, distribution, models, we should understand them very clearly in terms of what exactly they are.

I told you about parametric and non-parametric has the term parameter in between. If we assume anything about the parameter, that becomes parametric. If we do not assume about it, then it becomes non-parametric.

Just to help you understand the concept using a very simple example, on your board if you see I have two axes. Let us say the y-axis is sales of a company and the x-axis is advertisement on the budget that they placed. That is the amount of dollars that they spent.

Let us say the y-axis is the amount of money that they spent and the x-axis is the advertisement and the y-axis is the sales. Now if you plot the entire data, it might look like this. All of this could be a combination of advertisements and sales.

If you zoom into one specific data point here, let us say this one, it could be, let us say this particular data point on advertisement axis, it could be 100 dollars of spent. Similarly, the sales of the number of units of sales and let us say there are 200 units which are sold by spending 100 dollars. Now each of these data points which I am encircling right now, they are a combination of these two which are sales and advertisement.

So any data point can be represented as a combination of these two here. So this entire plot that you are seeing on front of your screen is called a scatter plot. What exactly is a scatter plot? Scatter plot is plot on which data points are scattered.

So you see every single data point is plotted and you are looking at them in a scattered way. They are all scattered here and there on the two-dimensional axis. Now if you see and if you would want to understand how does sales of a company or this company is related to the advertisement, you can sketch out a nice looking curve which says that sales is equals to two times of advertisements plus let us say 0.5. Now this 0.5 is a biased term which we add basically because if you think of if you spend nothing on the sales there still would be some amount of or let's say if you do not spend anything on the advertisement there still would be some sort of sales right.

So if you think of if you do not have no advertisement spent there would still be some sort of a sales here right and that's what we call as biased term. So let's say this is



the equation which relates to sales and advertisement. Now in this you may want to learn about these two parameters.

So I could write the same thing as y is equals to $ax + b$ wherein a is 2 and b is 0.5 right, y is sales and x is advertisement. The entire idea is to learn about these two parameters a and b right. Now how do we learn about this entire parameter is what the statistics or statistical learning is all about but because we are assuming that the relationship between y and x or sales and advertisement is represented or can be represented via this type of a linear relationship okay.

That's why we call this type of model or statistical learning method as parametric and what are the parameters involved in this you already know about sales you already know about advertisement but what you do not know about are these two terms a and b which are parameters. Now in this entire series of videos you will learn about different types of parametric methods. I was just helping you understand one of them but you would also learn about various different types of parametric methods parametric tests.

I just wanted to help you understand parametric as in whenever you have certain assumption about relationship about test about distribution those are called as parametric wherein the assumption is laid down about the shape of the curve shape of the relationship or the shape of the distribution and all of these things and then you have to infer about the parameter. Remember I told you about statistics all about taking a sample okay then assuming about the population using the sample inferring about the population using the sample itself that's what entire statistics or that's where the entire statistics revolves around right. So with that let's move into the next section and learn more about the other types of modules that we have to go through and we'll learn about different types of statistical learning methods different types of tests different types of modules and so on okay thank you

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